Health watch — main board

Sơ đồ phân cấp: mỗi ô là một trang con. Nhãn toàn cục (global label) nối tín hiệu giữa các trang; ký hiệu nguồn nối theo tên.

Ref 101–199. Sinh tự động bởi tools/gen_sch.py từ tools/design.py.

Power: USB-C, AXP2101 PMIC, battery, current shunts

File: health_watch_main_power.kicad_sch

ESP32-S3R8 + 16 MB flash + RF

File: health_watch_main_mcu.kicad_sch

Display FPC, IMU, RTC, LED boost, sensor-board FPC

File: health_watch_main_periph.kicad_sch

Motor, buzzer, SOS button, status LED

File: health_watch_main_ui.kicad_sch

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harisama1 — đồ án đồng hồ sức khỏe

Sheet: /
File: health_watch_main.kicad_sch

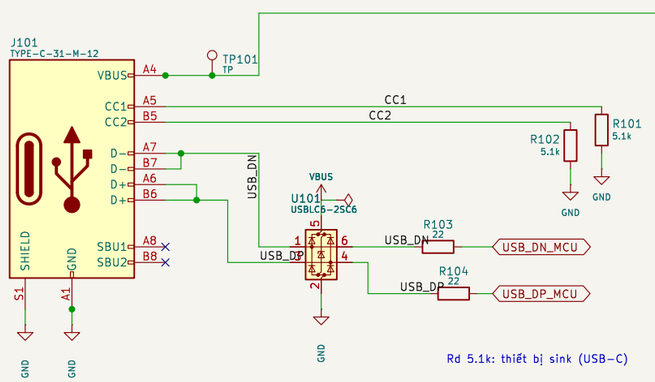
Title: Health watch — main board — root

Size: A4	Date: 2026-11-04	**Rev: 1.0**
KiCad E.D.A. 9.0.8		Id: 1/5

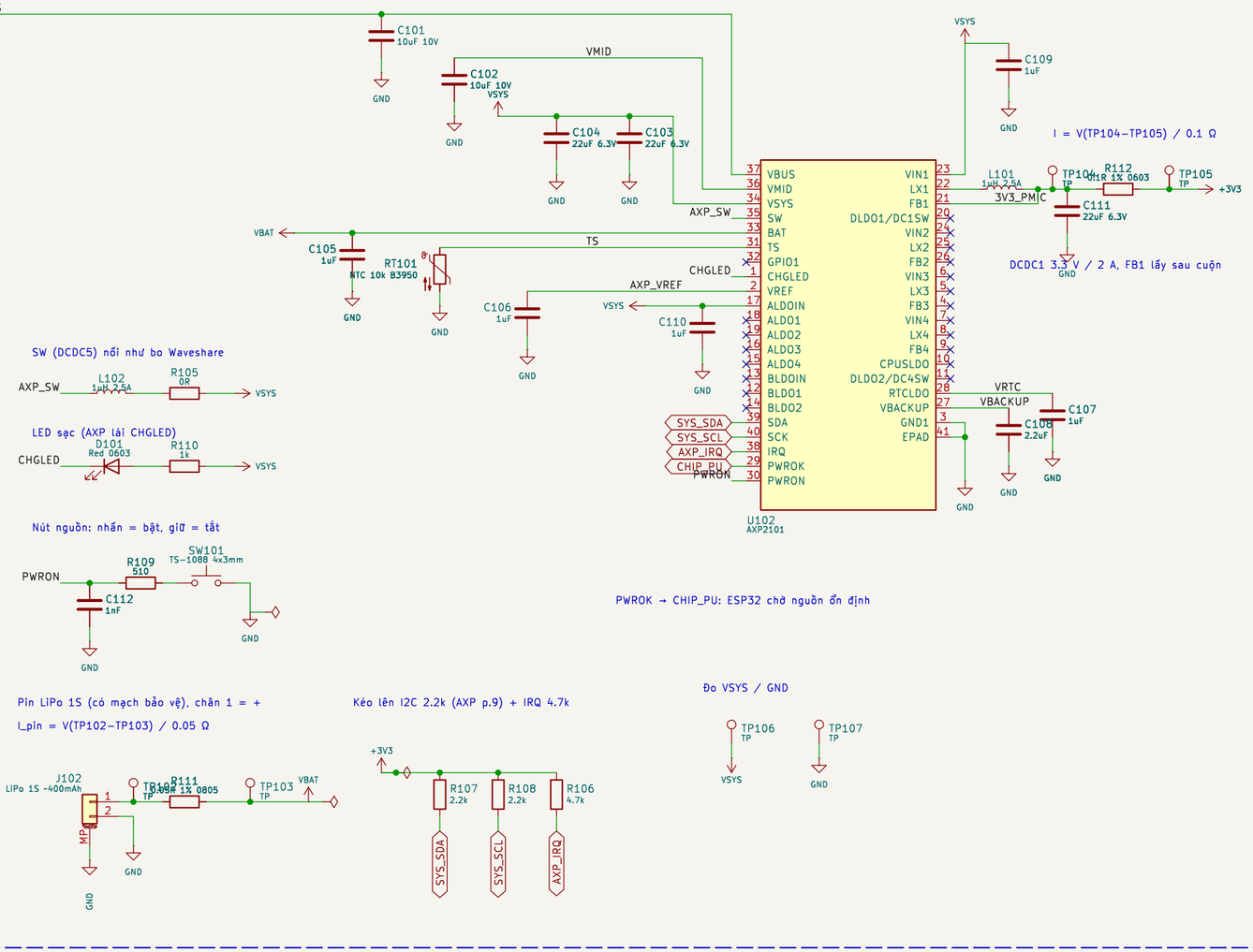
Power: USB-C, AXP2101 PMIC, battery, current shunts

USB-C → ESD → AXP2101 (sạc LiPo + DCDC1 3.3 V). Shunt + test point để đo dòng: pin (R111) và 3V3 (R112).

USB-C + ESD



PMIC AXP2101 + pin + shunt đo dòng



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Sheet: /Power: USB-C, AXP2101 PMIC, battery, current shunts/
File: health_watch_main_power.kicad_sch
Title: Health watch — main board — Power: USB-C, AXP2101 PMIC, battery,
Size: A3 | Date: 2026-11-04 | **Rev: 1.0**
KiCad E.D.A. 9.0.8 | Id: 2/5

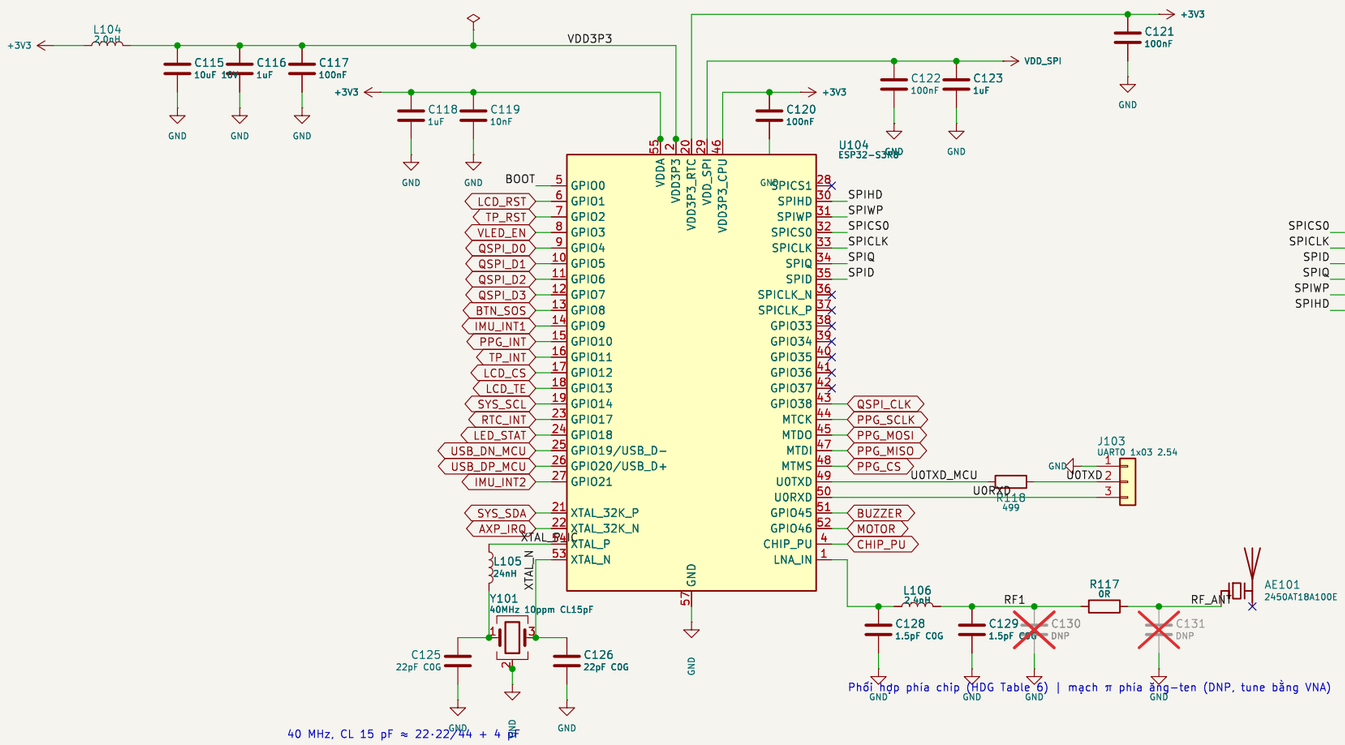
ESP32-S3R8 + 16 MB flash + RF

ESP32-S3R8: decoupling theo HDG Fig.1 p.5, thạch anh 40 MHz, flash QSPI 16 MB, mạch phối hợp RF + ăng-ten gốm, UART0.

ESP32-S3R8 + flash + thạch anh + RF + UART0

VDD_SPI = 3.3 V (GPIO45 = 0 lúc reset) cấp flash + PSRAM

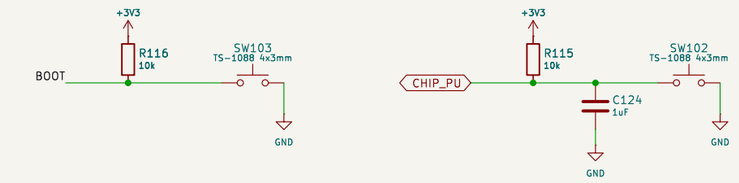
VDD3P3 (PA RF): lọc LC 2.0 nH



Phối hợp phía chip (HDG Table 6) | mạch π phía ăng-ten (DNP, tune bằng VNA)

40 MHz, CL 15 pF ≈ 22·22/44 + 4 pF

Reset (CHIP_PU) + Boot (GPIO0)



Giữ BOOT + nhấn RESET = chế độ nạp

RC 10k/1µF: CHIP_PU lên sau nguồn ≥ 50 µs (HDG §1.3.3)

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Sheet: /ESP32-S3R8 + 16 MB flash + RF/

File: health_watch_main_mcu.kicad_sch

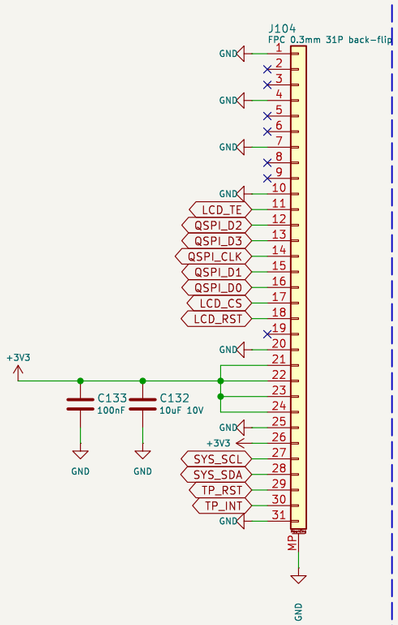
Title: Health watch — main board — ESP32-S3R8 + 16 MB flash + RF

Size: A3	Date: 2026-11-04	**Rev: 1.0**
KiCad E.D.A. 9.0.8		Id: 3/5

Display FPC, IMU, RTC, LED boost, sensor-board FPC

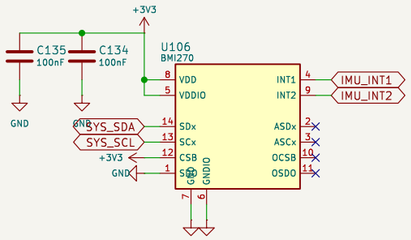
Màn hình AMOLED (FPC 31P như Waveshare 1.75C), IMU BMI270, RTC PCF85063A, boost 5 V cho LED PPG, FPC 16P sang bo cảm biến.

FPC màn hình AMOLED + cảm ứng

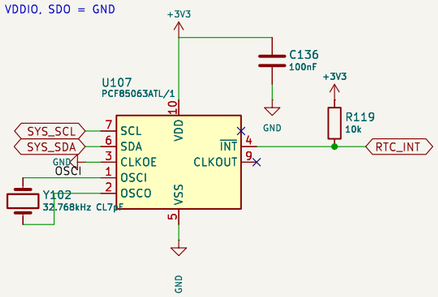


QSPI + I2C chung

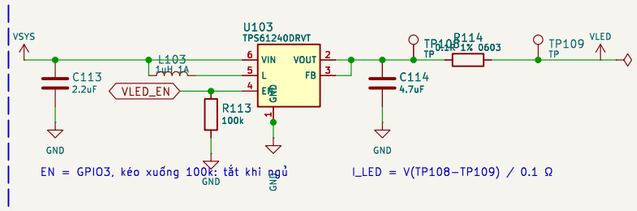
IMU BMI270 + RTC PCF85063A



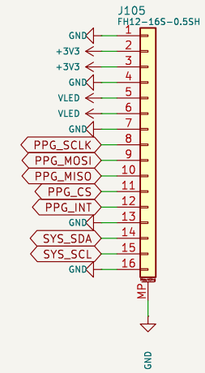
I2C 0x68: CSB = VDDIO, SDO = GND



Boost 5 V cho LED PPG (TPS61240)



FPC 16P sang bo cảm biến



Cáp FFC loại B: chân 1 ↔ chân 1

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Sheet: /Display FPC, IMU, RTC, LED boost, sensor-board FPC/
File: health_watch_main_periph.kicad_sch

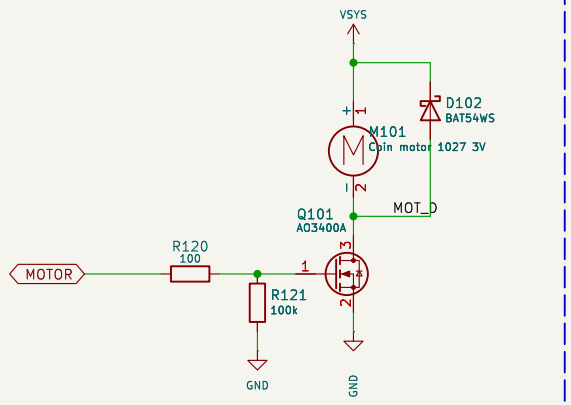
Title: Health watch — main board — Display FPC, IMU, RTC, LED boost, sensc

Size: A3	Date: 2026-11-04	**Rev: 1.0**
KiCad E.D.A. 9.0.8		Id: 4/5

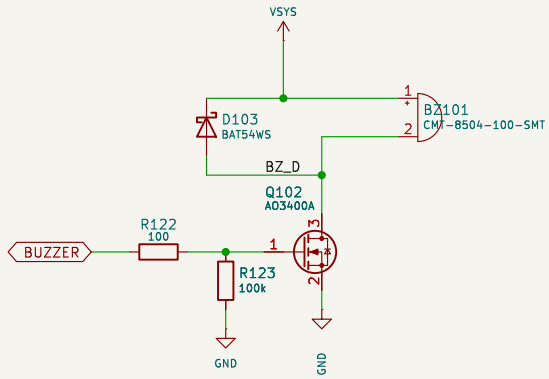
Motor, buzzer, SOS button, status LED

Motor rung + còi từ: MOSFET phía thấp, diode dập, cổng kéo xuống (GPIO45/46 là chân strapping). Nút SOS, LED trạng thái.

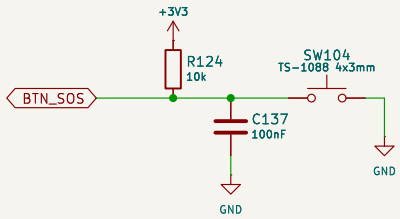
Motor rung (GPIO46)



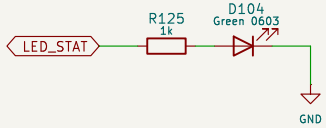
Còi từ (GPIO45, PWM ~2.7 kHz)



Nút SOS (GPIO8, đánh thức deep-sleep)



LED trạng thái (GPIO18)



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Sheet: /Motor, buzzer, SOS button, status LED/
File: health_watch_main_ui.kicad_sch

Title: Health watch — main board — Motor, buzzer, SOS button, status LED

Size: A4	Date: 2026-11-04	**Rev: 1.0**
KiCad E.D.A. 9.0.8		Id: 5/5